

Solutions to Mock JEE MAIN – 22 | JEE 2022

PHYSICS

1.(C) Heat loss $= \frac{1}{2} \frac{c_1 c_2}{c_1 + c_2} (v_1 + v_2)^2 = \frac{1}{2} \frac{8}{6} (900) \mu J = 600 \mu J$

2.(D) At height 'h' $|h \ll R|$

$$g = g_0 \left(1 - \frac{2h}{R} \right) \quad g_0 = \text{acceleration due to gravity at earth's surface}$$

$$\Rightarrow 0.97 g_0 = g_0 \left(1 - \frac{2h}{R} \right)$$

$$\therefore h = \frac{0.03R}{2}$$

At depth 'd'

$$g = g_0 \left(1 - \frac{d}{R} \right) \Rightarrow 0.99 g_0 = g_0 \left(1 - \frac{d}{R} \right)$$

$$\therefore d = 0.01R \Rightarrow \text{Ratio} = \frac{2}{3}$$

3.(D) $H_2 = 32.0 \text{ cm}$

$$H_1 = 22.0 \text{ cm}$$

$$\frac{\lambda}{2} = H_2 - H_1 = 32.0 - 22.0 = 10 \text{ cm}$$

$$\lambda = 20 \text{ cm} = 1/5 \text{ (m)}$$

$$v = f \lambda$$

$$v = 1615 \times \frac{1}{5} = 323 \text{ m/s}$$

4.(A) $S \propto \Delta p$

S depends upon v, f, ρ

$$\Rightarrow S = \Delta p \times v^x \rho^y f^z$$

$$\Rightarrow L = (ML^{-1}T^{-2})^1 (LT^{-1})^x (ML^{-3})^y (T^{-1})^z$$

$$\Rightarrow L = M^{1+y} L^{-1+x-3y} T^{-2-x-z}$$

$$\Rightarrow 1+y=0 \Rightarrow y=-1$$

$$-1+x-3y=1 \Rightarrow x=-1$$

$$-2-x-z=0 \Rightarrow z=-1$$

$$\Rightarrow S = \frac{\Delta p}{\rho v f} \text{ or } f = \frac{\Delta p}{\rho v S}$$

$$\Rightarrow f = \frac{100}{1 \times 330 \times 10 \times 10^{-3}} = \frac{1000}{33} \cong 30 \text{ Hz}$$

5.(B) Line current $= \frac{1100}{220} = 5A$

Line power loss $= I^2 R = 25 R$

Power supplied $= 1100 + 25R$

Efficiency $= \frac{\text{Power delivered}}{\text{Power supplied}}$

$0.9 = \frac{1100}{1100 + 25R} \Rightarrow (0.9)(25R) = 1100(1 - 0.9) \quad ; \quad R = \frac{1100}{225} = 4.9 \Omega$

6.(A) Image forms on object, when rays from object travels same path after reflection by mirror so rays refracted by lens, will form image at centre of curvature of mirror

By lens formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad ; \quad \frac{1}{v} + \frac{1}{60} = \frac{1}{30}$

$v = 60cm$

From lens distance of centre of curvature $= 60 \text{ cm}$

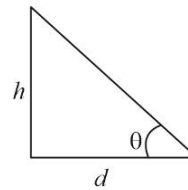
radius of curvature $= 30 \text{ cm}$

7.(C) At a general height h of balloon,

$\tan \theta = \frac{h}{d} \quad \therefore \quad \sec^2 \theta \frac{d\theta}{dt} = \frac{1}{d} \frac{dh}{dt}$

$\frac{d\theta}{dt} = \omega \text{ and } \frac{dh}{dt} = v$

$\therefore \quad \omega = \frac{v \cos^2 \theta}{d} = \frac{1}{30} \left(\frac{30}{50} \right)^2 = 0.012 \text{ rad/s}$



8.(D) Loss in electrostatic $PE = \frac{\sigma}{2\epsilon_0} (y) \times q$

Loss in gravitational $PE = mg y \Rightarrow \frac{1}{2} mv^2 = \frac{\sigma q}{2\epsilon_0} y + mg y \Rightarrow v^2 = 2y \left(\frac{q\sigma}{2m\epsilon_0} + g \right)$

9.(B) When a potential difference of 100V is applied across the voltmeter, full scale deflection should take place. Thus, $50\mu A$ should go through the coil. We add a resistance R in series with the given coil to achieve this. We have,

$50\mu A = \frac{100}{10^6 + R}$

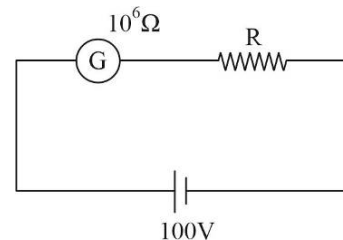
$\therefore R + 10^6 = 2 \times 10^6$

$R = 10^6 \Omega$

10.(B) $t = \frac{1}{\lambda} \ln \left(\frac{A_0}{A} \right) \& t_{mean} = \frac{1}{\lambda}$

From graph $165hrs = \frac{1}{\lambda} \ln \left(\frac{3 \times 10^{10}}{10^{10}} \right)$

$\frac{1}{\lambda} = \frac{165}{\ln 3} = \frac{165}{1.1} = 150hrs \quad ; \quad t_{mean} = \frac{1}{\lambda} = 150hrs$



- 11.(D) Angle between area vector and magnetic field = 90°

Torque due to magnetic force acting on coil

$$= n i A B \sin \theta = n i l^2 B$$

Torque due to weight = $mg(l/2)$

For the coil to start tipping

$$(Torque)_{due\ to\ \vec{B}} \geq (Torque)_{due\ to\ Mg}$$

$$\text{For minimum value of } \vec{B}, \quad \Rightarrow \quad n i l^2 B = mg \frac{l}{2} \Rightarrow B = \frac{mg}{2 n i l}$$

- 12.(C) The change in internal energy is same for AB, AC and AD and is positive

$$w_{AB} > 0 \ \& \ w_{AC} = 0 \ \& \ w_{AD} < 0$$

So by first law of thermodynamics

$$Q_{AB} > 0; \quad Q_{AC} > 0$$

For AD

$$Q_{AD} > 0 \text{ or } Q_{AD} < 0 \text{ both are possible}$$

13.(D) $Z = \frac{a^m c^p}{b^n d^q}$

$$\frac{\Delta Z}{Z} = m \frac{\Delta a}{a} + n \frac{\Delta b}{b} + p \frac{\Delta c}{c} + q \frac{\Delta d}{d}$$

$$\text{Given, } 9 = m \left(\frac{1}{2} \right) + n \left(\frac{3}{2} \right) + p \left(\frac{5}{2} \right) + q \left(\frac{7}{2} \right) \quad \therefore m + 3n + 5p + 7q = 18$$

- 14.(B) By the principle of calorimetry

$$200S_P(90 - 120) + 100S_Q(90 - 75) = 0 \quad \dots(i)$$

$$\text{and } 100S_P(T - 100) + 50S_Q(T - 70) = 0 \quad \dots(ii)$$

where S_P & S_Q are specific heats of P and Q respectively and T is the final temperature in second case.

Solving (i), we get

$$4S_P = S_Q$$

Solving (ii), we get

$$\frac{S_P}{S_Q} = \frac{-(T - 70)}{2(T - 100)}$$

$$\text{Thus, } \frac{1}{4} = \frac{-(T - 70)}{2(T - 100)}; \quad T - 100 = -2T + 140$$

$$T = 80^\circ\text{C}$$

- 15.(B) When V becomes half of its initial value

$$\therefore \text{ pressure becomes 4 times } (V^2 P = \text{constant})$$

So, temperature becomes 2T ($PV/T = \text{constant}$)

Given,

$$E = \frac{f}{2} K(2T) \quad \left| \quad \begin{array}{l} E = 5.44 \times 10^{-17} \text{ ergs} \\ f = 3 \end{array} \right.$$

$$\therefore T = \frac{E}{f K} \quad \left| \quad \begin{array}{l} K = \text{Boltzman constant} \end{array} \right.$$

$$\text{Now, } \frac{N}{N_A} \times R \times T = PV$$

$$N = \frac{PVf}{E} = \frac{13.6 \times 980 \times 4 \times 5 \times 3}{5.44 \times 10^{-17}} = 1.47 \times 10^{22}$$

16.(D) Let density of oil, ball and water be ρ_0 , ρ_b & ρ_w respectively.

$$\text{Then, } \rho_0 = \frac{\rho_w}{3}; \quad \rho_b = \frac{2}{3}\rho_w$$

Further, assume that volume of ball is V , out of which V' volume is submerged in water.

As the ball is in equilibrium, gravitational force is balanced by upward buoyant force

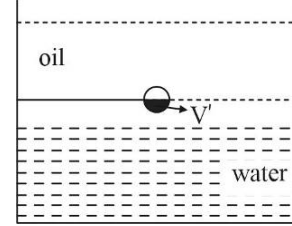
$$F_g = F_{\text{buoyant} | \text{oil}} + F_{\text{buoyant} | \text{water}}$$

$$(\rho_b V)g = \rho_0(V - V')g + \rho_w(V')g$$

$$\rho_b V = \rho_0 V - \rho_0 V' + \rho_w V'$$

$$(\rho_b - \rho_0)V = (\rho_w - \rho_0)V'$$

$$V' = \left(\frac{\rho_b - \rho_0}{\rho_w - \rho_0} \right) V$$



$$\text{Substituting value of } \rho_b \text{ \& } \rho_0 \text{ in terms of } \rho_w, \quad V' = \left(\frac{\frac{\rho_w}{3}}{\frac{2}{3}\rho_w} \right) V \Rightarrow V' = \frac{V}{2}$$

$$\% \text{ volume of ball submerged in water} = \frac{V'}{V} \times 100 = 50\%$$

$$17.(D) \quad E_g = \frac{hc}{\lambda_{\max}} = \frac{12400}{\lambda_m(\text{in } \text{\AA})} eV = \frac{12400}{2500} eV = 4.96 eV$$

$$18.(D) \quad \text{Time taken to cross first 81\% of fall} \Rightarrow \sqrt{\frac{2(0.81)H}{g}} \Rightarrow 0.9\sqrt{\frac{2H}{g}}$$

$$\text{Time taken to fall completely} \Rightarrow \sqrt{\frac{2H}{g}}$$

$$\therefore \text{ time taken to cross last 19\% of fall} \Rightarrow \sqrt{\frac{2H}{g}} - 0.9\sqrt{\frac{2H}{g}} \Rightarrow 0.1\sqrt{\frac{2H}{g}} \Rightarrow \frac{1}{10}\sqrt{\frac{2H}{g}}$$

$$19.(A) \quad \vec{E} = 90 \hat{j} (0.3 \times 10^7 t - 10^{-2} x) V/m.$$

$$\hat{C} = \hat{i}$$

$$\hat{B} = \hat{C} \times \hat{E} = \hat{i} \times \hat{j} = \hat{k}; \quad B_0 = \frac{E_0}{C} = \frac{90}{3 \times 10^8} = 3 \times 10^{-7} \text{ Tesla}$$

$$\vec{B} = 3 \times 10^{-7} \hat{k} (0.3 \times 10^7 t - 10^{-2} x)$$

$$F_{B_{\max}} = qvB_0 = (2 \times 1.6 \times 10^{-19}) \times 0.01 \times 3 \times 10^8 \times 3 \times 10^{-7} = 3.2 \times 9 \times 10^{-19-2+8-7} = 2.88 \times 10^{-19} N$$

20.(C) The angular momentum of two wheels and shaft is constant

$$I\omega = (3I + I)\omega'$$

$$\omega' = \frac{\omega}{4} \quad \text{So impulse on first wheel shaft system} = \frac{I\omega}{4} - I\omega = \frac{-3I\omega}{4}$$

21.(8) $\phi = \vec{B} \cdot \vec{A}$
 $\phi = BA \cos \theta$

$A = \text{Area of triangle} = \frac{\sqrt{3}a^2}{4}$; $\phi = B_0 e^{-\lambda t} \frac{\sqrt{3}a^2}{4} \times (1)$

$\text{emf} = \frac{-d\phi}{dt} = \frac{B_0 \lambda e^{-\lambda t} \sqrt{3}a^2}{4}$; $I = \frac{\text{emf}}{3R} = \frac{B_0 \lambda e^{-\lambda t} \sqrt{3}a^2}{4 \times 3R} = \frac{B_0 e^{-\frac{t}{2}} a^2}{2 \times 4\sqrt{3}R}$

$x = 8$

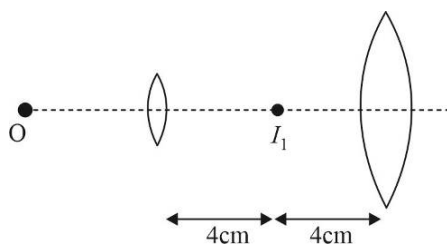
22.(12) For least strain I_1 from objective should be formed at focus of the eyepiece

$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_0}$

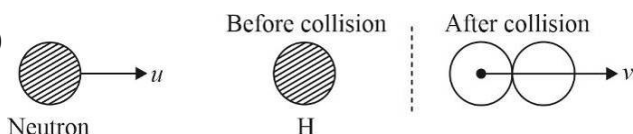
$\frac{1}{u_1} = \frac{1}{v_1} - \frac{1}{f_0}$

$\frac{1}{u_1} = \frac{-1}{4} \Rightarrow u_1 = -4\text{cm}$

$\Rightarrow$ object should be placed 4cm from the objective and 12cm from the eyepiece.



23.(6.80)



Neutron

H

Both neutron & hydrogen will have the same KE.

($\because$ They have same mass & same velocity)

By conservation of linear momentum

$mu = 2mv \Rightarrow v = u/2$

$\Rightarrow$ KE of neutron after collision $k' = \frac{1}{2}m(u/2)^2 = \frac{k}{4}$

Energy lost in collision $= k - \left(\frac{k}{4} + \frac{k}{4}\right) = k/2$

$k/2 = 13.6\text{eV}$ for hydrogen to get ionized $\Rightarrow k = 27.2\text{eV}$ & $k' = 6.8\text{eV}$

24.(3) $\vec{\tau} = \vec{r} \times \vec{F}$ or, $\tau = rF \sin \theta$

$3\sqrt{3} = 6 \times 1 \times \sin \theta$; $\sin \theta = \frac{\sqrt{3}}{2}$; $\theta = 60^\circ$ or $\frac{\pi}{3}$ Thus, $n = 3$

25.(55) $\lambda =$ De-broglie wavelength for electron of kinetic energy KE

$\lambda = \frac{h}{\sqrt{2mk}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9 \times 10^{-31} \times 5 \times 1.6 \times 10^{-19}}} = \frac{6.6 \times 10^{-34}}{3 \times 4 \times 10^{-25}} = 5.5 \times 10^{-10}\text{m}$

Successive localized accumulation are location of constructive interference

$\Rightarrow$ Separation between them is fringe width (β)

$\beta = \frac{\lambda D}{d} \Rightarrow \beta = \frac{5.5 \times 10^{-10} \times 1}{10 \times 10^{-6}} = 5.5 \times 10^{-5}\text{m} = 55\mu\text{m}$

26.(4) Total equivalent resistance $= \frac{4 \times 4}{4 + 4} + 1 + \frac{12 \times 6}{12 + 6} + 1 = 8\Omega$

Total current in circuit $= \frac{V}{R_{eq}} = \frac{16}{8} = 2\text{A}$; $V_{AB} = IR_{AB} = 2 \times 2 = 4\text{V}$

27.(0.75) Compass needles oscillates in horizontal plane due to horizontal component of magnetic field H

$$T = 2\pi\sqrt{\frac{I}{MH}}, H = B\cos\delta$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{B_2 \cos\delta_2}{B_1 \cos\delta_1}} \Rightarrow \frac{B_1}{B_2} = \left(\frac{T_2}{T_1}\right)^2 \frac{\cos\delta_2}{\cos\delta_1}$$

$$T_1 = \frac{1}{30} \text{ min}, T_2 = \frac{1}{40} \text{ min} \Rightarrow \frac{T_2}{T_1} = \frac{3}{4}$$

$$\frac{B_1}{B_2} = \left(\frac{3}{4}\right)^2 \frac{\cos 37^\circ}{\cos 53^\circ} = \left(\frac{3}{4}\right)^2 \frac{4}{3} = \frac{3}{4} = 0.75$$

28.(11) Angle between the mirrors = $10 + 20 = 30^\circ$

$$\Rightarrow \frac{360}{30} = 12 \Rightarrow \text{Even number} \quad \therefore \text{Total number of images} = \frac{360}{\theta} - 1 = 12 - 1 = 11$$

29.(15) Speed just before reaching B is given by energy conservation

$$mg(5) = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(10)(5)} = 10 \text{ m/s}$$

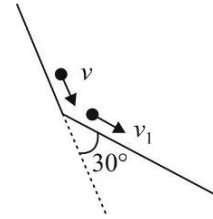
After collision at B, velocity perpendicular to the incline becomes zero while velocity along the incline remains unchanged

$$\therefore v_1 = v \cos 30^\circ = 5\sqrt{3} \text{ m/s}$$

Velocity upon reaching C can be found by applying energy conservation again.

$$mg(7.5) = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Rightarrow v_2^2 = v_1^2 + 2g(7.5) = 75 + 150 = 225 \quad \therefore v_2 = 15 \text{ m/s}$$



30.(0.18) Area of face parallel to which force is applied = $50\text{cm} \times 10\text{cm} = 0.05\text{m}^2$

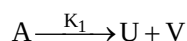
$$\therefore \text{The stress applied} = \frac{9 \times 10^4}{0.05} = 1.80 \times 10^6 \text{ Nm}^2$$

$$\text{Shearing strain} = \frac{\Delta x}{L} = \frac{\text{stress}}{G}$$

$$\therefore \text{displacement } (\Delta x) = \text{stress} \times \frac{L}{G} = \frac{1.8 \times 10^6 \times 0.5}{5 \times 10^9} = 1.8 \times 10^{-4} \text{ m} = 0.18 \text{ mm}$$

Chemistry

1.(B) For reactions:-



50% of reaction is completed,
time required for 50% completion

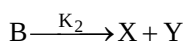
$$= (t_{1/2})_A = t(\text{let say}) \dots (i)$$

From equation (i) and (ii), we get

$$(t_{1/2})_A = 3(t_{1/2})_B$$

$$\frac{\ln 2}{K_1} = \frac{3 \ln 2}{K_2} \left[\because (t_{1/2})_A = \frac{\ln 2}{K_1}, (t_{1/2})_B = \frac{3 \ln 2}{K_2} \right]$$

$$K_2 = 3K_1 \quad \text{or} \quad \frac{K_2}{K_1} = 3$$



87.5% of reaction is completed,
time required for 87.5% completion =

$$3(t_{1/2})_B = t \dots (ii)$$

2.(B) The basicity strength will be directly proportional to the availability of lone pair on N atom.

3.(A) (I) Becomes aromatic, when H^+ is removed

(II) Becomes aromatic, when H^+ is removed

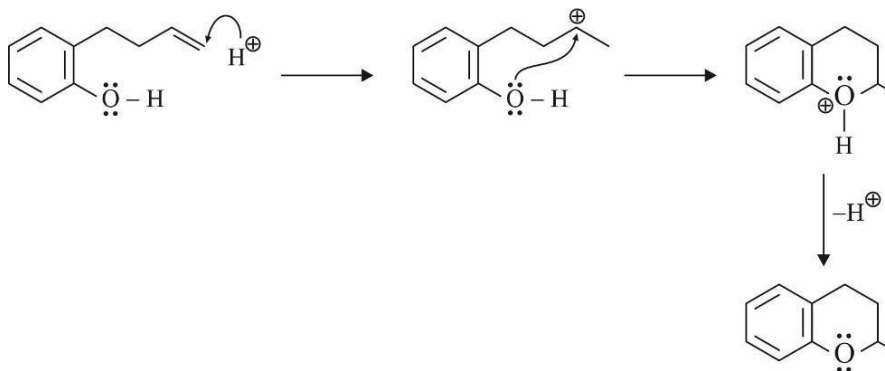
(In comparison to (I), In (II) an EWG ($-NO_2$) is attached, which makes hydrogen even more acidic)

(III) Becomes anti-aromatic, when H^+ is removed

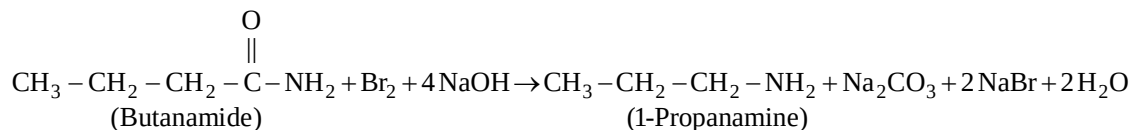
Hence, K_a (or acidic strength order) : II > I > III

So, pK_a order: II < I < III [Increasing order] or III > I > II [decreasing order]

4.(C)



5.(D) Hoffmann bromamide degradation reaction



6.(C) Lysine is basic as well as essential amino acid.

7.(D) In the given Ellingham diagram, below 1500 K, the $C \rightarrow CO$ curve lie below the $Fe \rightarrow FeO$ curve. Hence, conversion of $C \rightarrow CO$ is more favorable. Hence, carbon can reduce FeO .

8.(D) Micelles are formed above a certain concentration called critical micelle concentration (CMC) and above a particular temperature called kraft temperature (T_k)

9.(D) Presence of excess of nitrate in drinking water causes Blue baby syndrome.

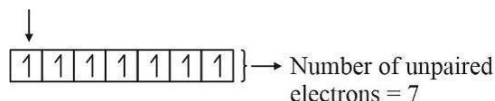
10.(A) Electronic configuration of $_{58}\text{Ce} : [\text{Xe}]4f^1 5d^1 6s^2$

The most common oxidation state of Ce is +3 as cerium is a lanthanoid.

But Ce can show +4 oxidation state also it is f^0 in +4.

11.(D) $_{65}\text{Tb} : [\text{Xe}]4f^9 6s^2$

$\text{Tb}^{4+} : [\text{Xe}] 4f^7$ (correct electronic configuration)



$$\mu = \sqrt{n(n+2)} \quad (\text{where } n = \text{Number of unpaired electrons})$$

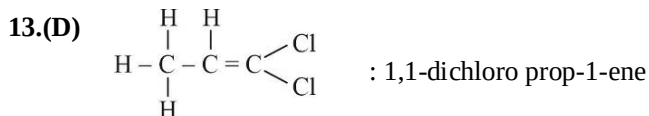
$$\downarrow$$

$$\text{spin-only} = \sqrt{7(7+2)} = 7.9$$

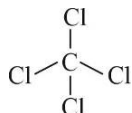
magnetic

moment

12.(C) s orbitals can not form π bonds. s orbitals always form σ bonds.



sp^2 hybridization = 120° bond angle between $\text{Cl}-\text{C}-\text{Cl}$



sp^3 hybridization = 109° bond angle between $\text{Cl}-\text{C}-\text{Cl}$

Thus "D" is false

14.(C) $\mu = \sqrt{n(n+2)} \text{ BM}$

$$4.89 = \sqrt{n(n+2)}$$

$$n = 4$$

Thus the answer will have 4 unpaired electrons in its electronic configuration

(I) $\text{Fe} : [\text{Ar}] 4s^2 3d^6$

It has 4 unpaired electrons in d subshell

(II) $\text{Sc} : [\text{Ar}] 4s^2 3d^1$

It has only 1 unpaired electron

(III) $\text{Fe}^{2+} : [\text{Ar}] 3d^6$

It has 4 unpaired electrons

(IV) $\text{Cr}^{2+} : [\text{Ar}] 3d^4$

It has 4 unpaired electrons

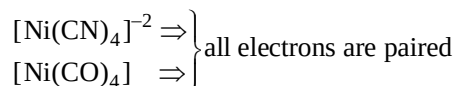
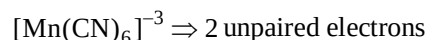
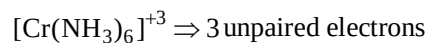
Thus answer is I, III and IV

- 15.(A)** According Le-chatelier principle, when pressure is increased, reaction shift in a direction in which number of moles of gases decrease. Thus reaction will shift in forward direction

Since, this is an exothermic reaction, decreasing temperature will shifts reaction in forward direction

When inert gas is added at constant pressure, reactions shifts in a direction in which number of moles of gases will increase. Thus reaction will shifts in backward direction

- 16.(D)** Diamagnetic species will not be attracted by magnetic field



CO and CN^- are strong field ligands. Hence, pairing occurs

- 17.(B)** In FCC, $4r = a\sqrt{2} \Rightarrow a = \frac{4r}{\sqrt{2}}$

$$\therefore a = \frac{4}{\sqrt{2}} \times 90\sqrt{2} = 360 \text{ pm}$$

$$= 360 \times 10^{-10} \text{ cm}$$

$$= 3.6 \times 10^{-8} \text{ cm}$$

And effective number of atoms per unit cell is 4.

$$\text{Density}(d) = \frac{n \times M_0}{N_A \times a^3}$$

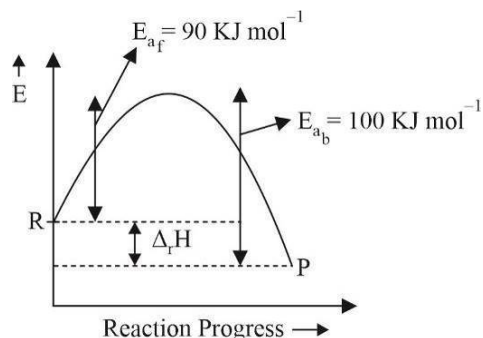
$$M_0 = \frac{d \times N_A \times a^3}{n}$$

$$M_0 = \frac{8 \times 6.023 \times 10^{23} \times (3.6 \times 10^{-8})^3}{4}$$

$$M_0 = 56.2 \approx 56$$

- 18.(C)** Type E_2O_3 oxides of Nitrogen & phosphorus are purely acidic. That of arsenic & antimony amphoteric and those of bismuth are predominantly basic.
- 19.(D)** Solubility decreases with increase in size of alkyl/aryl groups
- 20.(C)** Synthetic drugs, brompheniramine (Dimetapp) and terfenadine (Seldane), act as antihistamines. They interfere with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.

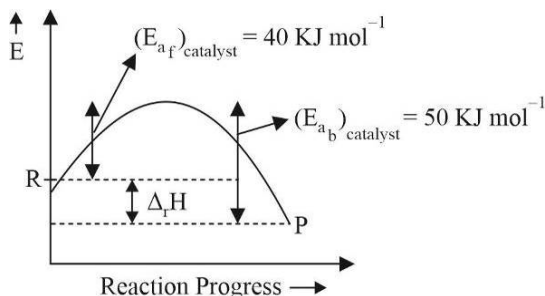
21.(10) In absence of catalyst :



$$\Delta_r H = -(E_{ab} - E_{af})$$

$$\Delta_r H = -10 \text{ kJ mol}^{-1}$$

In presence of catalyst :



$$\Delta_r H = -(E_{ab} - E_{af})_{\text{catalyst}}$$

$$\Delta_r H = -10 \text{ kJ mol}^{-1}$$

On, addition of catalyst, $\Delta_r H$ doesn't change, only activation energies are lowered or reduced.

So, $|\Delta_r H|$ (in kJ) = 10 kJ mol^{-1}

22.(23.16) Balanced reaction : $\text{Zn(s)} + 2\text{MnO}_2\text{(s)} + 8\text{H}^+ \rightarrow \text{Zn}^{2+} + 2\text{Mn}^{3+} + 4\text{H}_2\text{O}$

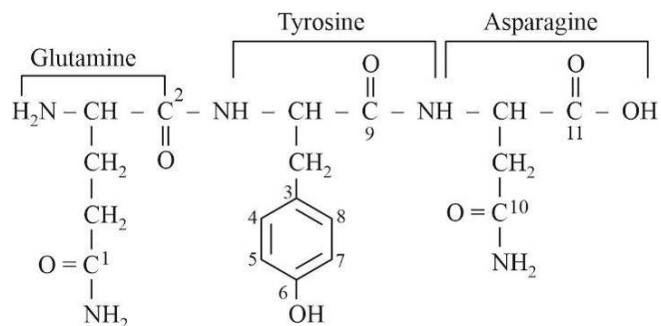
For the above reaction : $n = 2$ [no. of electrons involved]

$$\text{and, } E_{\text{cell}}^{\circ} = (E_{\text{oxidation}}^{\circ})_{\text{anode}} + (E_{\text{Reduction}}^{\circ})_{\text{cathode}}$$

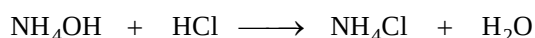
$$E_{\text{cell}}^{\circ} = 1.2\text{V}$$

$$\begin{aligned} \text{Using } \Delta G^{\circ} &= -nFE_{\text{cell}}^{\circ} = -2 \times 96500 \times 1.2 \\ &= -2.4 \times 96500 \\ &= -231600 \text{ Joules} \end{aligned}$$

$$\text{So, } |\Delta G^{\circ}| \times 10^{-4} = 23.16$$

23.(11) Number of sp^2 hybridized carbon = 1124.(18) millimoles of $\text{NH}_4\text{OH} = 20 \times 0.2 = 4$

millimoles of $\text{HCl} = 20 \times 0.1 = 2$



Initially 4 2 - -

Finally 2 0 2 -

Let us add x millimole of HCl

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$K_b = 10^{-5} \Rightarrow pK_b = 5,$$

$$6 = 5 + \log \frac{(2+x)}{(2-x)}$$

$$1 = \log \frac{2+x}{2-x}$$

$$10 = \frac{2+x}{2-x}$$

$$20 - 10x = 2 + x$$

$$18 = 11x$$

$$\frac{18}{11} = x = M \times V$$

$$\frac{18}{11} = \frac{1}{11} \times V$$

$$V = 18\text{ml}$$

$$\begin{aligned} 25.(10) \quad \text{no. of moles of } N_2 \text{ evolved} &= \frac{\text{volume evolved}}{\text{molar volume}} \\ &= \frac{448}{22400} = 0.02 \end{aligned}$$

$$\text{Mass of } N = 0.02 \times 28 = 0.56\text{g}$$

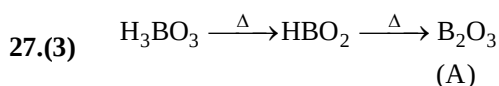
$$\% \text{ of } N = \frac{0.56}{5.6} \times 100 = 10\%$$

$$\begin{aligned} 26.(40) \quad \Delta T_b &= i K_b m & \Delta T_b &= 5.04 \\ i &= \frac{\Delta T_b}{K_b m} & K_b &= 2.52 \\ i &= \frac{5.04}{2.52 \times 2.5} & m &= \frac{200/40}{2} \\ & & m &= 2.5 \end{aligned}$$

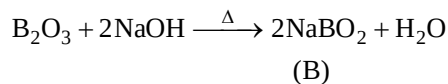
$$i = \frac{2}{2.5} = 0.8$$

$$\text{In association, } \alpha = \frac{1-i}{1-\frac{1}{n}} = \frac{1-0.8}{1-\frac{1}{2}} = \frac{0.2}{0.5} = 0.4$$

$$\Rightarrow \text{So, } 100\alpha = 0.4 \times 100 = 40$$



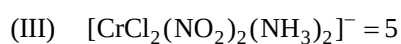
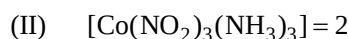
Boron trioxide reacts with NaOH to form NaBO₂ (sodium metaborate) and water.



In above reaction, 2 moles of NaBO₂ requires 1 mole of B₂O₃

∴ for 6 moles of NaBO₂, 3 moles of B₂O₃ will be required

28.(34) Possible number of geometrical isomers



$$\therefore x = 1; y = 2; z = 5$$

$$2(xy + yz + xz) = 2(2 + 10 + 5) = 34$$

29.(12) $\Delta_t = \frac{4}{9} \Delta_0$

$$\Delta_t \text{ for } [\text{ML}_4]^{-2} = \frac{4}{9} \times 27000$$

$$= 12000$$

$$= 12 \times 10^3 \text{ cm}^{-1}$$

$$\therefore x = 12$$

30.(39.20) mass of H₂SO₄ present in 1 ml of stock solution = $1.5 \times \frac{50}{100} = 0.75 \text{ g}$

$$\text{mass of H}_2\text{SO}_4 \text{ required for 750 ml of 0.4M H}_2\text{SO}_4 = \frac{750}{1000} \times 0.4 \times 98 = 29.4 \text{ g}$$

1 ml stock solution contained 0.75g H₂SO₄.

By unitary method

$$29.4 \text{ g H}_2\text{SO}_4 \text{ will be in } \frac{29.4}{0.75} \text{ ml} = 39.2 \text{ ml}$$

Mathematics

- 1.(B) The tangent to the parabola $y^2 = 8x$ with slope ' M ' is $y = Mx + \frac{2}{M}$

It is also tangent to hyperbola $xy = -1$

$$x\left(Mx + \frac{2}{M}\right) = -1 \text{ have equal roots}$$

$$Mx^2 + \frac{2}{M}x + 1 = 0$$

$$D = 0 \Rightarrow \frac{4}{M^2} = 4M \Rightarrow M^3 - 1 = 0$$

$M = 1$ (other roots are imaginary)

$\therefore$ required equation is $y = x + 2$

- 2.(A) Let $\vec{a}, \vec{b}, \vec{c}$ are 3 coterminal edge of parallelepiped ; then $V = [\vec{a} \vec{b} \vec{c}]$

Diagonal will be $\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}$
 $d_1 \quad d_2 \quad d_3$

$$[\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}] = 2[\vec{a} \vec{b} \vec{c}] = [\vec{d}_1 \vec{d}_2 \vec{d}_3]$$

$$V = \frac{1}{2}[\vec{d}_1 \vec{d}_2 \vec{d}_3] = \frac{1}{2}[\hat{i} + \hat{j} \quad \hat{j} + \hat{k} \quad \hat{k} + \hat{i}] = \frac{1}{2} \times 2[\hat{i} \hat{j} \hat{k}] = 1$$

- 3.(B) $49\% = n(B) = \%$ of passengers who liked Bus for travelling
 $53\% = n(F) = \%$ of passengers who liked Flight for travelling
 $62\% = n(T) = \%$ of passengers who liked Train for travelling
 $n(B \cap F) = 27\% ; n(F \cap T) = 29\% ; n(B \cap T) = 28\%$

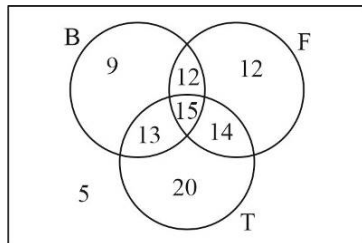
$\therefore 5\%$ liked none $\Rightarrow$ so, $n(B \cup F \cup T) = 95\%$

Using formulas

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$95\% = (49 + 53 + 62)\% - 27\% - 29\% - 28\% + n(B \cap F \cap T)$$

$$n(B \cap F \cap T) = 15\%$$



Passengers who liked at least two mode of transport

$$= (12 + 13 + 14 + 15)\% = 54\% \text{ of } 200$$

$$= \frac{54}{100} \times 200 = 108$$

4.(A) Let Q be the image of Pt. $P(0, -1, -3)$ in the plane $3x - y + 4z = 2$

PQ is normal to plane

$\therefore$ Direction ratios of $PQ : (3, -1, 4)$

$\therefore PQ$ passes through $(0, -1, -3)$

Equation of PQ :

$$\frac{x-0}{3} = \frac{y+1}{-1} = \frac{z+3}{4} = \lambda \text{ (say)}$$

Coordinates of Q be $(3\lambda, -\lambda-1, 4\lambda-3)$

Let (R) be mid pt. of PQ

R lies in plane $3x - y + 4z = 2$

$$R : \left(\frac{3\lambda}{2}, \frac{-\lambda-2}{2}, \frac{4\lambda-6}{2} \right)$$

$$3 \times \frac{3\lambda}{2} + \frac{\lambda+2}{2} + 2(4\lambda-6) = 2$$

$$9\lambda + \lambda + 2 + 16\lambda - 24 = 4$$

$$26\lambda = 26 \Rightarrow \lambda = 1 \quad \therefore Q : (3, -2, 1)$$

5.(C) The converse statement of $(x \wedge (\sim y)) \Rightarrow k$ is $k \Rightarrow (x \wedge (\sim y))$

$$\begin{aligned} &= \sim k \vee (x \wedge \sim y) \\ &= (x \wedge \sim y) \vee (\sim k) \\ &= [\sim (\sim x \vee y)] \vee (\sim k) \\ &= (\sim x \vee y) \Rightarrow (\sim k) \end{aligned} \quad \left| \begin{array}{l} (\because p \Rightarrow q \equiv \sim p \vee q) \\ \therefore (x \wedge \sim y \equiv \sim (\sim x \vee y)) \end{array} \right.$$

$\therefore$ Converse of $(x \wedge (\sim y)) \Rightarrow k$ is $(\sim x \vee y) \Rightarrow (\sim k)$

$$6.(B) \quad \sum_{i=1}^n (x_i + 1)^2 = 9n \Rightarrow \sum_{i=1}^n x_i^2 + 2 \sum_{i=1}^n x_i + n = 9n \quad \dots(i)$$

$$\sum_{i=1}^n (x_i - 1)^2 = 5n \Rightarrow \sum_{i=1}^n x_i^2 - 2 \sum_{i=1}^n x_i + n = 5n \quad \dots(ii)$$

Add (i) + (ii)

$$2 \sum_{i=1}^n x_i^2 = 12n$$

$$\sum_{i=1}^n x_i^2 = 6n$$

Subtract (ii) from (i)

$$\sum_{i=1}^n x_i = n$$

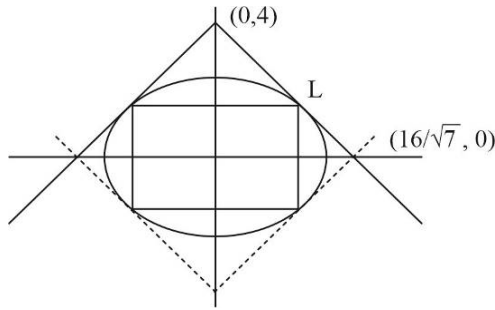
$$SD = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2} = \sqrt{\frac{6n}{n} - \left(\frac{n}{n} \right)^2} = \sqrt{6-1} = \sqrt{5}$$

7.(A) $9x^2 + 16y^2 = 144 \Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$

$$a = 4, b = 3 \Rightarrow e^2 = \frac{a^2 - b^2}{a^2} = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$$

Coordinates of one end of LR in 1st quadrant are

$$L\left(ae, \frac{b^2}{a}\right) \text{ i.e., } L\left(\sqrt{7}, \frac{9}{4}\right) \Rightarrow \left(\sqrt{7}, \frac{9}{4}\right)$$



Equation of tangent at L is $\sqrt{7}x + 4y = 16$

Point of intersection with axes are $P\left(\frac{16}{\sqrt{7}}, 0\right)$ and $(0, 4)$

$$\text{Area of quad. formed} = 4 \times \frac{1}{2} \times \frac{16}{\sqrt{7}} \times 4 = \frac{128}{\sqrt{7}}$$

8.(C) $y_2(x^2 + 1) = 2xy_1$

$$\int \frac{y_2}{y_1} = \int \frac{2x}{x^2 + 1}$$

$$\log y_1 = \log(x^2 + 1) + C$$

$$y_1|_{x=0} = 2$$

$$\log 2 = \log(1) + C \Rightarrow C = \log 2$$

$$\log y_1 = \log(x^2 + 1) + \log(2) = \log(2x^2 + 2)$$

$$\frac{dy}{dx} = 2x^2 + 2$$

$$\int dy = \int (2x^2 + 2) dx$$

$$y = 2 \times \frac{x^3}{3} + 2x + K \rightarrow \text{passes through } (3, 0)$$

$$0 = 2 \times \frac{3^3}{3} + 6 + K \Rightarrow K = -24$$

$$\text{Equation of curve : } y = \frac{2x^3}{3} + 2x - 24$$

9.(C) $x + y + z = 6$
 $x + 2y + 3z = 10$
 $x + 2y + \lambda z = \mu$

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = (\lambda - 3)$$

$$D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 3 \\ \mu & 2 & \lambda \end{vmatrix} = 2\lambda - 16 + \mu$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 10 & 3 \\ 1 & \mu & \lambda \end{vmatrix} = 2(2\lambda - \mu + 4)$$

$$D_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 10 \\ 1 & 2 & \mu \end{vmatrix} = \mu - 10$$

For $\lambda = 3, \mu = 10 \rightarrow$ option C

$$D = 0 \text{ \& } D_1 = D_2 = D_3 = 0$$

Hence given system has infinite solution

10.(B) $f(\theta) = \begin{vmatrix} 1 & 4\cos\theta & 1 \\ \sin\theta & 1 & 4\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$

Applying $R_3 \rightarrow R_3 - R_1$

$$f(\theta) = \begin{vmatrix} 1 & 4\cos\theta & 1 \\ \sin\theta & 1 & 4\cos\theta \\ 0 & (\sin\theta - 4\cos\theta) & 0 \end{vmatrix}$$

$$\begin{aligned} & 1(0 - 4\cos\theta(\sin\theta - 4\cos\theta)) - 4\cos\theta(0 - 0) + 1(\sin\theta(\sin\theta - 4\cos\theta)) \\ &= -4\cos\theta(\sin\theta - 4\cos\theta) + \sin\theta(\sin\theta - 4\cos\theta) \\ &= (\sin\theta - 4\cos\theta)(\sin\theta - 4\cos\theta) = (\sin\theta - 4\cos\theta)^2 \end{aligned}$$

$$\sin\theta - 4\cos\theta \leq \sqrt{(1)^2 + (-4)^2}$$

$$\sin\theta - 4\cos\theta \leq \sqrt{17}$$

$$(\sin\theta - 4\cos\theta)^2 \leq 17$$

11.(A) $f(g(x)) = x$ (given $f \circ g = I$)

$$f'(g(x))g'(x) = 1 \quad \forall x$$

$$f'(g(a))g'(a) = 1$$

$$f'(g(a)) = \frac{1}{g'(a)} = \frac{1}{2}$$

$$f'(b) = \frac{1}{2}$$

12.(C) $z^2 - z + 1 = 0$

$$z = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2} = \frac{1 \pm \sqrt{3}i}{2} = -\omega_1, -\omega^2$$

Put $z = -\omega$

$$\left(-\omega + \frac{1}{-\omega}\right)^2 + \left(\omega^2 + \frac{1}{\omega^2}\right)^2 + \left(-\omega^3 + \frac{1}{-\omega^3}\right)^2 + \dots + \left((- \omega)^{24} + \frac{1}{(-\omega)^{24}}\right)^2$$

$$\left(-\omega - \frac{\omega^2}{\omega^3}\right)^2 + \left(\omega^2 + \frac{\omega}{\omega^3}\right)^2 + (-1-1)^2 + \left(\omega^4 + \frac{1}{\omega^4}\right)^2 \dots + \left(\omega^{24} + \frac{1}{\omega^{24}}\right)^2$$

$$\underbrace{(1)^2 + (-1)^2 + (2)^2 + (\omega + \omega^2)^2 + \dots + (1+1)^2}_{=6}$$

The sum 6 is repeated in 8 triplets $\therefore$ Answer is $6 \times 8 = 48$

13.(B) $\lim_{x \rightarrow p} (1 + ax^2 + bx + c)^{\frac{1}{x-p}}$

This is the form $(\rightarrow 1)^{\rightarrow \infty}$

$$= \lim_{x \rightarrow p} (1 + ax^2 + bx + c - 1) \cdot \frac{1}{x-p}$$

$$= \lim_{x \rightarrow p} \frac{(a(x-p)(x-q))}{(x-p)} \quad [\text{since } p, q \text{ are roots of } ax^2 + bx + c]$$

$$= e^{a(p-q)}$$

14.(B) $\cos^{-1} x + \cos^{-1} \left(x \times \frac{1}{2} + \sqrt{1-x^2} \cdot \sqrt{1-\frac{1}{4}} \right)$

$$= \cos^{-1} x + \left(\cos^{-1} \frac{1}{2} - \cos^{-1} x \right)$$

$$= \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$$

$$\therefore, \cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1} \left(xy + \sqrt{1-x^2} \sqrt{1-y^2} \right); & x < y, x, y > 0 \\ -\cos^{-1} \left(xy + \sqrt{1-x^2} \sqrt{1-y^2} \right); & x > y, x, y > 0 \end{cases}$$

15.(C) $f(x) = px^2 + qx + r$

$$f'(x) = 2px + q$$

$$f(1) = f(-1)$$

$$p + q + r = p - q + r \Rightarrow q = 0$$

$$f'(p) = 2p^2 + q = 2p^2$$

$$f'(q) = 2pq + q = 0$$

$$f'(r) = 2pr + q = 2pr$$

Given $f'(p), f'(q), f'(r)$ are in A.P.

$$\Rightarrow 2p^2, 0, 2pr \text{ are in A.P.}$$

$$\Rightarrow \frac{2p^2}{2p}, 0, \frac{2rp}{2p} \text{ are in A.P.} \Rightarrow p, 0, r \text{ are in A.P.}$$

i.e., p, q, r are in A.P.

16.(B) $P \equiv (P_1 \cos \theta, P_1 \sin \theta)$

$$Q \equiv (P_2 \cos(90 - \theta), -P_2 \sin(90 - \theta))$$

$$Q \equiv (P_2 \sin \theta, -P_2 \cos \theta)$$

Put coordinates of P in $y^2 = 4ax$

$$P_1^2 \sin^2 \theta = 4aP_1 \cos \theta$$

$$P_1 = \frac{4a \cos \theta}{\sin^2 \theta}$$

Similarly putting coordinates of Q in $y^2 = 4ax$

$$P_2 = \frac{4a \sin \theta}{\cos^2 \theta}$$

$$(P_1 \cdot P_2)^{4/3} = \left(\frac{4a \cos \theta}{\sin^2 \theta} \times \frac{4a \sin \theta}{\cos^2 \theta} \right)^{4/3}; (P_1 \cdot P_2)^{4/3} = \frac{(16a^2)^{4/3}}{(\sin \theta \cdot \cos \theta)^{4/3}}$$

$$\text{Also, } P_1^{2/3} + P_2^{2/3} = \left(\frac{4a \cos \theta}{\sin^2 \theta} \right)^{2/3} + \left(\frac{4a \sin \theta}{\cos^2 \theta} \right)^{2/3}$$

$$= (4a)^{2/3} \left(\frac{\cos^{2/3} \theta}{\sin^{4/3} \theta} + \frac{\sin^{2/3} \theta}{\cos^{4/3} \theta} \right) = (4a)^{2/3} \left[\frac{\cos^2 \theta + \sin^2 \theta}{\sin^{4/3} \theta \cos^{4/3} \theta} \right] = (4a)^{2/3} \left[\frac{1}{\sin^{4/3} \theta \cos^{4/3} \theta} \right]$$

$$\text{Also, } 16a^2 (P_1^{2/3} + P_2^{2/3}) = \frac{16a^2 \times (4a)^{2/3}}{\sin^{4/3} \theta \cos^{4/3} \theta} = \frac{(16a^2)^{4/3}}{\sin^{4/3} \theta \cos^{4/3} \theta} = \frac{(16a^2)^{4/3}}{(\sin \theta \cos \theta)^{4/3}} = (P_1 \cdot P_2)^{4/3}$$

$\therefore$ option (B) is correct

17.(A) $a_1 = \frac{1}{2 \times 5} = \frac{1}{3} \left(\frac{1}{2} - \frac{1}{5} \right)$

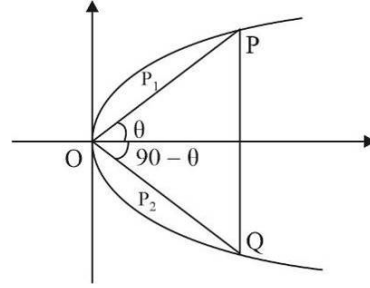
$$a_2 = \frac{1}{5 \times 8} = \frac{1}{3} \left(\frac{1}{5} - \frac{1}{8} \right)$$

$\vdots$
 $\vdots$
 $\vdots$

$$a_{100} = \frac{1}{299 \times (302)} = \frac{1}{3} \left(\frac{1}{299} - \frac{1}{302} \right)$$

$$a_1 + a_2 + \dots + a_{100} = \frac{1}{3} \left[\frac{1}{2} - \frac{1}{5} + \frac{1}{5} - \frac{1}{8} + \dots + \frac{1}{299} - \frac{1}{302} \right]$$

$$= \frac{1}{3} \left[\frac{1}{2} - \frac{1}{302} \right] = \frac{1}{3} \left[\frac{300}{2 \times 302} \right] = \frac{50}{302} = \frac{25}{151}$$



$$18.(D) \quad |6x^2 - 5x + 1| = 5x - 6x^2 - 1$$

$$= -(6x^2 - 5x + 1)$$

$$\text{Here } |a| = -a$$

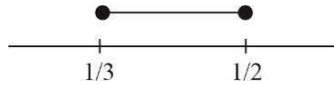
$$\Rightarrow a \leq 0$$

$$6x^2 - 5x + 1 \leq 0$$

$$6x^2 - 3x - 2x + 1 \leq 0$$

$$3x(2x - 1) - 1(2x - 1) \leq 0$$

$$(2x - 1)(3x - 1) \leq 0$$



$$x \in \left[\frac{1}{3}, \frac{1}{2} \right]$$

$$19.(B) \quad I = \int_0^{80\pi} \frac{dx}{1 + e^{\tan x}} \quad \dots(i)$$

$$I = \int_0^{80\pi} \frac{dx}{1 + e^{\tan(80\pi - x)}}$$

$$= \int_0^{80\pi} \frac{dx}{1 + e^{-\tan x}} \quad \dots(ii)$$

Add (i) and (ii)

$$2I = \int_0^{80\pi} \left(\frac{1}{1 + e^{\tan x}} + \frac{1}{1 + e^{-\tan x}} \right) dx$$

$$= \int_0^{80\pi} \left(\frac{1}{1 + e^{\tan x}} + \frac{e^{\tan x}}{1 + e^{\tan x}} \right) dx$$

$$= \int_0^{80\pi} 1 dx$$

$$= 80\pi$$

$$2I = 80\pi$$

$$I = 40\pi$$

$$20.(D) \quad \int x \log \left(1 + \frac{1}{x} \right) dx = I$$

$$I = \int x \log \left(\frac{1+x}{x} \right) dx$$

$$I = \int \left(\underbrace{x \log(1+x)}_{I_1} - \underbrace{x \log x}_{I_2} \right) dx$$

$$\begin{aligned}
 I_1 &= \int x \log(1+x) dx \\
 \text{using integration by parts} \\
 \log(1+x) \cdot \frac{x^2}{2} - \int \frac{1}{1+x} \cdot \frac{x^2}{2} dx &= \frac{x^2}{2} \cdot \log x - \int \frac{x}{2} dx \\
 \frac{x^2}{2} \log(1+x) - \frac{1}{2} \int \frac{x^2}{1+x} dx &= \frac{x^2}{2} \log x - \frac{x^2}{4} \\
 I &= I_1 - I_2 \\
 I &= \frac{x^2}{2} \log(1+x) - \frac{1}{2} \int \frac{x^2-1+1}{1+x} dx - \frac{x^2}{2} \log x + \frac{x^2}{4} \\
 &= \frac{x^2}{2} \log(1+x) - \frac{1}{2} \int (x-1) dx - \frac{1}{2} \int \frac{1}{1+x} dx - \frac{x^2}{2} \log x + \frac{x^2}{4} + C \\
 &= \frac{x^2}{2} \log(1+x) - \frac{x^2}{4} + \frac{x}{2} - \frac{1}{2} \log(1+x) - \frac{x^2}{2} \log x + \frac{x^2}{4} + C \\
 &= \left(\frac{x^2-1}{2} \right) \log(1+x) - \frac{x^2}{2} \log x + \frac{x}{2} + C = \left(\frac{x^2-1}{2} \right) \log(1+x) - \frac{x^2}{4} \log x^2 + \frac{x}{2} + C \\
 &= f(x) \log(1+x) + g(x) \log x^2 + Lx + C \\
 f(x) &= \frac{x^2-1}{2}, \quad g(x) = \frac{-x^2}{4}, \quad L = \frac{1}{2}
 \end{aligned}$$

- 21.(2454) We have 11 letters in 'VIDYAMANDIR'
 2 I's, 2 A's, 2 D's and V, Y, M, N, R are single
 Case – I :- Two alike & other two alike

$$\text{No. of ways} = {}^3C_2 \times \frac{4!}{2! \times 2!}$$

Case – II :- Two alike, two different

$$\text{No. of ways} = {}^3C_1 \times {}^7C_2 \times \frac{4!}{2!}$$

Case – III :- All are different

$$\text{No. of ways} = {}^8C_4 \times 4!$$

$$\therefore \text{Total no. of ways} = {}^3C_2 \cdot \frac{4!}{2! \times 2!} + {}^3C_1 \cdot {}^7C_2 \times \frac{4!}{2!} + {}^8C_4 \times 4! = 18 + 756 + 1680 = 2454$$

- 22.(4) General term $T_{r+1} = K_{C_r} (1)^{K-r} (x^2)^r$

$$T_{r+1} = K_{C_r} x^{2r}$$

For $x^4, r = 2$

$$K_{C_2} = 6$$

$$\frac{K!}{2!(K-2)!} = 6$$

$$\frac{K(K-1)(K-2)!}{2(K-2)!} = 6$$

$$K = -3 \text{ or } K = 4$$

$\therefore$ positive value of K is 4.

23.(5) x^2 is continuous, $\therefore 2x^2$ is also continuous

Range of $2x^2$ in $[1, 2)$ will be $2 \leq 2x^2 < 8$

G.I.F. is discontinuous at integers

$f(x)$ will be discontinuous at integer in $[2, 8)$

$$f(1) = f(1^+) = 2 \text{ continuous at } x=1$$

No. of points of discontinuity is $5 \Rightarrow \sqrt{\frac{3}{2}}, \sqrt{2}, \sqrt{\frac{5}{2}}, \sqrt{3}, \sqrt{\frac{7}{2}}$

24.(5) Possibility of getting 7 are

(1,6) (6,1) (2,5) (5,2) (3,4) (4,3)

$$\text{Probability of getting score of 7 in a single throw} = \frac{6}{36} = \frac{1}{6}$$

$$\therefore \text{Required probability} = \text{probability of getting score of 7 exactly twice} = {}^3C_2 \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)$$

$$= \frac{3!}{2!(3-2)!} \frac{1}{36} \times \frac{5}{6} = \frac{3}{36} \times \frac{5}{6} = \frac{5}{72}$$

$$\therefore 72P = \frac{5}{72} \times 72 = 5$$

25.(1) $4x - 2y + 6 = 0$

$$4x - 2y + 8 = 0$$

$$\alpha = \frac{|8-6|}{\sqrt{16+4}} = \frac{2}{\sqrt{20}} = \frac{1}{\sqrt{5}}$$

$$6x - 3y + 9 = 0$$

$$6x - 3y + 15 = 0$$

$$\beta = \frac{|15-9|}{\sqrt{36+9}} = \frac{6}{\sqrt{45}} = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\therefore \alpha^2 + \beta^2 = \frac{1}{5} + \frac{4}{5} = 1$$

26.(1) $y = (x-2)^2 + 1$

$$x-2 = \pm\sqrt{y-1}$$

$$\therefore x = -\sqrt{y-1} + 2 \quad (\because x \leq 2)$$

Interchange x & y

$$\therefore y = f^{-1}(x) = -\sqrt{x-1} + 2$$

$\therefore$ domain of $f^{-1}(x)$ is $[1, \infty)$

$$\therefore a = 1$$

$$27.(13) \quad f(x) = \sqrt{x+9}$$

$$f'(x) = \frac{1}{2\sqrt{x+9}}$$

$$f'(C) = \frac{f(b) - f(a)}{b - a} \Rightarrow \frac{1}{2\sqrt{C+9}} = \frac{\sqrt{7+9} - \sqrt{0+9}}{7-0} \Rightarrow \frac{1}{2\sqrt{C+9}} = \frac{1}{7}$$

$$\frac{7}{2} = \sqrt{C+9} ; \frac{49}{4} = C+9 ; C = \frac{49}{4} - 9 = \frac{49-36}{4} = \frac{13}{4}$$

$$28.(4) \quad 2ab^2 - 4ab = 6b^2$$

$$ab = 2a + 3b$$

$$(a-3)b = 2a$$

$$b = \frac{2a}{a-3}$$

$$\therefore ab = \frac{2a^2}{a-3} = k$$

$$\frac{dk}{da} = \frac{2(a^2 - 6a)}{(a-3)^2} = 0$$

$$\therefore a = 0, 6$$

$$\text{At } a = 6, \frac{dk^2}{da^2} = +ve$$

$$\therefore \text{ when } a = 6 \text{ \& } b = 4, ab \text{ is minimum} \quad \therefore \text{ minimum } \frac{ab}{6} = \frac{6 \times 4}{6} = 4$$

$$29.(38) \quad \text{Volume} = |(A \times B) \cdot C|$$

$$= \begin{vmatrix} 3 & 4 & 5 \\ 5 & 6 & 1 \\ 0 & 2 & 3 \end{vmatrix} = 3(6 \times 3 - 1 \times 2) - 4(5 \times 3 - 1 \times 0) + 5(5 \times 2 - 6 \times 0) = 48 - 60 + 50 = 38$$

$$30.(7) \quad \text{Area} = \int_0^1 (\sqrt{x} - x^4) dx$$

$$= \left[\frac{2}{3} x^{3/2} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{5} - (0 - 0)$$

$$k = \frac{7}{15}$$

$$\therefore 15k = 7$$

